

Problem Set 2: Consumer Theory, Demand & Welfare

Microeconomic Theory for Experimentalists & Applied Microeconomists

Due: [date — before Lecture 1 of Module 3]

Ground rules: collaboration on ideas is encouraged; write-ups are individual. You may use AI tools for parts flagged with ⟨AI⟩ and must attach the transcript. Show all calculations.

Weights: Part A — 30 points. Part B — 45 points. Part C — 25 points.

Part A: Theory — Guided Calculations (30 points)

Question 1 (Demand and the Slutsky decomposition — 15 points). *A consumer has utility $u(x, y) = xy$, income $m = 24$, and faces prices $p_y = 3$ throughout.*

- (a) **(4 pts)** *At the optimum, the marginal rate of substitution y/x equals the price ratio p_x/p_y . Use this and the budget constraint to show that demand is $x = \frac{m}{2p_x}$ and $y = \frac{m}{2p_y}$ — i.e. she spends half her income on each good. Compute the bundle at $p_x = 3$.*
- (b) **(3 pts)** *The price of x falls to $p_x = 2$. Compute the new bundle and state the total effect on x .*
- (c) **(5 pts)** *Slutsky decomposition. Compute the compensated income m' that makes the original bundle exactly affordable at the new prices. Compute demand for x at $(p_x = 2, m')$, and report the substitution and income effects separately. Hint: $m' = m - x_{\text{old}} \cdot \Delta p_x$ where Δp_x is the price DROP; both effects here are whole numbers.*
- (d) **(3 pts)** *A development agency is choosing between a price subsidy on good x and an equivalent-cost cash transfer (equal total budgetary outlay). In one paragraph, use your decomposition to explain what an experiment that measures only the total demand response to the subsidy cannot tell the agency, and what treatment arm fixes it.*

Question 2 (CV, EV, and consumer surplus with numbers — 15 points). *A consumer has quasi-linear utility: her marginal value for good x is $20 - 2x$ dollars, and utility is linear in money spent on everything else (so there are no income effects on x). The price of x rises from \$4 to \$8.*

- (a) **(3 pts)** *Compute her demand for x at each price. Hint: she buys until marginal value equals price.*
- (b) **(5 pts)** *Compute the loss in consumer surplus from the price rise (a trapezoid — draw it). Explain in one sentence why, with quasi-linear utility, CV and EV both equal this number.*
- (c) **(4 pts)** *Now suppose x were instead a normal good with income effects. State which is larger in absolute value for this price rise — CV or EV — and give the intuition in two sentences. Hint: CV asks the compensation question at NEW prices, when the consumer is poorer in real terms.*

- (d) (**3 pts**) A ministry must budget compensation payments for households hurt by this price rise, and separately wants to predict how households would vote on a referendum to block it. Which welfare measure belongs to which task, and why?

Part B: Experimental Design (45 points)

Question 3 (Is the WTP–WTA gap an endowment effect?). *Kahneman, Knetsch & Thaler (1990) found median WTA roughly twice median WTP for the same mug. The loss-aversion reading says this is a real feature of preferences. The skeptical reading (Plott–Zeisel style) says much of it is subject confusion about the elicitation mechanism and strategic habits imported from everyday bargaining. Design a lab experiment that measures the WTP–WTA gap AND separates these two readings. Specify:*

- (a) (**6 pts**) Subject pool and recruitment, and why prior experience with the mechanism (or its absence) matters for your question.
- (b) (**12 pts**) Treatments. Your design must include (i) an endowment manipulation (sellers vs. buyers of the same good) and (ii) at least one arm that changes subjects’ understanding or experience of the mechanism (e.g. paid practice rounds on induced-value tokens) without changing the good or the endowment. Say what each comparison identifies.
- (c) (**9 pts**) The elicitation mechanism and its incentive-compatibility argument, written for a skeptic in plain language (this is the tutor-session drill). Then name one failure mode the mechanism does NOT fix.
- (d) (**9 pts**) Hypotheses and rough sample-size reasoning: state the WTA/WTP ratio you expect in each arm under each reading (anchor on KKT’s ≈ 2 and on the training-shrinks-the-gap literature), and the approximate n per arm to distinguish them. Simulation-based reasoning is welcome; formulas are not required.
- (e) (**9 pts**) The biggest confound remaining in YOUR design and your answer to it.

Part C: Silicon Pilot Interpretation ⟨AI⟩ (25 points)

Question 4 (The agents that were too rational). You run `module2-simulation-lab.ipynb`. Across all personas and framings, your agents’ four-menu choices come back **100% WARP-consistent** — not a single violation in 180 responses. Human subjects in comparable budget-choice experiments (Andreoni & Miller 2002; Choi et al. 2007) are highly but imperfectly consistent: roughly one in ten violates GARP, and consistency scores vary meaningfully across people. (The comparison is qualitative: the lab’s audit is pairwise WARP over four menus, while the human statistics are GARP rates over longer menu sequences.) Your lab partner reports: “the agents replicate rational consumer theory — great news.”

- (a) (**8 pts**) Explain why perfect consistency is a divergence from human behavior, not a replication — and give two mechanisms (at least one LLM-specific) that could produce it.
- (b) (**9 pts**) Design the diagnostic simulation that tests whether silicon consistency is robust or fragile: propose at least two manipulations (e.g. non-round prices, more menus, distractor text in the prompt, a persona described as busy or distracted) and state what result pattern

under each manipulation would tell you. Run it if API budget allows and report; otherwise state the predicted patterns precisely.

- (c) **(8 pts)** *Your research group wants to use this lab to pilot a budget-choice experiment for human subjects — including choosing the number of menus and estimating how many subjects will fail rationality screens. State one thing the silicon pilot CAN safely inform, one thing it CANNOT (and why), and the design change this implies for the human pilot plan.*

Attach your AI transcripts and, if you ran code, the results CSV.